BEFORE THE FEDERAL COMMUNICATIONS COMMISSION

WASHINGTON, D.C. 20554

In the Matter of	)
Space Exploration Holdings, LLC	) ICFS File No.
Application for Authorization of	) SAT-LOA-20260108-00016
Non-Geostationary Orbit Satellite	)
System (Orbital Data Center)	)

INFORMAL OBJECTION

AND REQUEST FOR THRESHOLD RELIEF

OF NICKOLAI G. BAKKEN

Pursuant to 47 C.F.R. §§ 1.41 and 25.154(b)

SUMMARY OF THE FILING

This Informal Objection asks the Federal Communications Commission to resolve a set of threshold questions before processing Space Exploration Holdings, LLC's application for up to one million orbital data center satellites as ordinary Part 25 business. The Application's own record reveals a six-order-of-magnitude discrepancy between the encoded technical instrument and the described system, an admission that Schedule S "cannot accurately describe" the proposal, and additional apparent omissions bearing on evaluability, reviewability, and the public-interest determination. As filed, the record does not appear to contain a complete technical description of the proposed system at the scale described, submitted ITU coordination materials, a bounded lifecycle or environmental throughput envelope, an identified AI-governance framework, a clearly independent command pathway, or any described independent audit or verification regime.

The Objector asks the Commission to state, on the record, how it intends to evaluate, supervise, and if necessary enforce an authorization where the Applicant itself admits that the Commission's standard technical instrument cannot accurately describe the system proposed, and where multiple threshold predicates for ordinary processing appear unresolved.

This filing also places on the record post-filing developments that may bear on the completeness of the administrative record and the Commission's disposition of the Application, including changes to the Applicant's ownership and corporate structure after public notice, together with additional federal developments arising during the compressed thirty-day pleading window. The Objector requests that the Commission identify the rule-bound path it is using, explain whether the present record is sufficient for action, and make express findings on the threshold issues

raised herein. To the extent any such issues are not resolved in the Objector's favor, they are preserved to the fullest extent available for any later administrative or judicial proceeding.

Relief Requested

The Objector respectfully requests two alternative forms of relief.

First, procedural relief. The Commission should extend or reopen the pleading period, or otherwise require supplementation and renewed public notice, to the extent it concludes that material post-filing developments or record omissions altered the posture the public was asked to evaluate. This includes, at minimum, any supplementation necessary to address the current ownership and control picture, the technical record actually required for review of the system described, and any other material deficiency identified in this filing.

Second, if procedural relief is denied, deferred, or deemed unnecessary, merits relief on the present record. In that event, the Commission should:

(1) return the Application as unacceptable for filing under the applicable rules, or otherwise dismiss it without prejudice, if the record is not sufficiently complete, evaluable, and reviewable to support ordinary processing;

(2) if the Commission does not return or dismiss the Application, hold the matter in abeyance and require supplementation, clarification, or amendment sufficient to resolve the threshold defects identified herein;

(3) if substantial and material factual questions remain, designate the matter for hearing under 47 U.S.C. Section 309(e); and

(4) whatever disposition the Commission selects, make express findings on the threshold questions raised in this filing, including the procedural and legal basis for the path chosen.

The Objector further requests that whatever disposition the Commission selects, whether extension, reopening, renewed notice, return, dismissal without prejudice, abeyance, supplementation, or hearing designation, the Commission make findings sufficient to render its action reviewable under the Administrative Procedure Act. Threshold issues placed on the record should not be left to silent or implied resolution. See Motor Vehicle Mfrs. Ass'n v. State Farm Mut. Auto. Ins. Co., 463 U.S. 29, 43 (1983).

All issues not fully resolved are preserved to the fullest extent available for reconsideration, further administrative proceedings, and judicial review.

I. NATURE OF THE FILING

Nickolai G. Bakken, a United States citizen residing in Risaralda, Colombia, respectfully submits this Informal Objection under 47 C.F.R. Section 1.41.

This filing asks the Commission to address a set of threshold questions that appear on the face of the current record before the Application is processed as ordinary Part 25 business.

Objector submits this filing to preserve threshold defects, record omissions, and public-interest concerns bearing on the Commission's disposition of the application.

The Communications Act requires the Commission to act in the public interest, convenience, and necessity. On the present record, the Commission cannot comfortably make that finding because the Application appears to present an unprecedented orbital compute and AI infrastructure proposal at civilizational scale while material technical, ownership, and governance questions remain unresolved. The Commission should not grant the application unless and until those threshold questions are addressed on the record and the public has had a meaningful opportunity to test them.

The Objector files in good faith and is not a competitor in orbital matters. The request here is simple. If the record is sufficient, the Commission should say why. If the record is not sufficient, the Commission should say so and identify the lawful path forward. If the Commission relies on waiver, supplementation, consultation, or some other procedural mechanism, it should identify that mechanism on the record.

Objector submits this filing as an informal objection under 47 C.F.R. §§ 1.41 and 25.154(b). To the extent party-in-interest or standing concepts later become relevant in any subsequent administrative or judicial proceeding, Objector expressly preserves them.

A. Good-Faith Basis, Petition Right, and Public Record Sources

Objector proceeds pro se and submits this filing in good faith, based on the Objector's own review of the publicly available administrative record and other cited public materials. This filing is submitted in exercise of the right to petition the Government for a redress of grievances protected by the First Amendment, a right that extends to access to administrative agencies and courts. Objector submits this Informal Objection through the procedural vehicle provided by 47 C.F.R. § 1.41.

Where this filing references legal standards, scientific findings, technical issues, or reported federal developments, it does so by citing published authorities, enacted statutes, peer-reviewed research, official materials, or contemporaneous reporting identified in the filing. The Objector takes responsibility for the decision to file and for the selection and presentation of the issues raised herein, and respectfully requests that the Commission address the substance of the threshold questions presented rather than disregard them on account of pro se status. Courts hold pro se pleadings to less stringent standards than formal pleadings drafted by lawyers, and while this filing does not assert that the Commission is mechanically bound to apply judicial pleading doctrine in the same manner, the Objector respectfully submits that comparable solicitude is appropriate in evaluating a good-faith informal objection under 47 C.F.R. § 1.41.

To the extent any deficiency in this filing is deemed curable, the Objector respectfully requests leave to supplement or correct it within a reasonable period rather than having substantive issues disregarded on procedural grounds.

B. Waiver of Deficiencies

To the extent this filing does not conform to any requirement of the Commission's rules regarding format, form, length, citation style, or other procedural convention, the Objector respectfully requests waiver pursuant to 47 C.F.R. § 1.3. The Commission's rules may be waived for good cause shown, in whole or in part, at any time.

Good cause exists here. The Commission accepted for filing an Application containing multiple waiver requests, a six-order-of-magnitude discrepancy between its encoded and described systems, and no submitted ITU coordination materials, and moved it to public notice within five days. The resulting thirty-day pleading window for what the Applicant describes as a civilizational-scale undertaking was further compressed by multiple convergent federal developments, several arising in the final days before the deadline. In these circumstances, and given the unusual scale and apparent novelty of the issues presented, the Objector respectfully requests that any non-substantive procedural defect in this filing be waived or, if curable, addressed through leave to supplement rather than through disregard of the issues raised.

C. Record Documents

The Application record before the Commission consists of the following principal documents, which are referenced in this filing by the abbreviations indicated: the FCC Form 312 Application (Form 312); the Application Narrative (NAR); the Technical Attachment (TECH); the Schedule

S structured data submission (SCHS); the Waiver Requests (WAIV); and the Ownership Attachment (OWN). The Objector has reviewed these documents as publicly available through the Commission's International Communications Filing System, FCC Schedule S Instructions (SCHS-INST) Edition Date August 2024, OMB Control Number 3060-0678, approved under the Paperwork Reduction Act of 1995, 44 U.S.C. Sections 3501 through 3521.

II. THRESHOLD QUESTIONS THE COMMISSION SHOULD ANSWER

A. Completeness and Evaluability

The first question is whether this Application is sufficiently complete and evaluable, in the form presented, to permit ordinary processing at all.

The current record raises that question on its face. The Schedule S instrument encodes three satellites, three total satellites deployed during the license term, on three orbital planes, while the Narrative describes a system of up to one million satellites, a six-order-of-magnitude difference. NAR at 1; SCHS. The Waiver Requests further acknowledge that Schedule S "cannot accurately describe" the proposed system. WAIV at 4. Where the applicant itself states that the Commission's standard technical instrument cannot accurately describe the system for which authorization is sought, a threshold adequacy problem exists under 47 C.F.R. Section 25.112(a).

The same concern appears in the channel description. A range of one to one hundred dynamic channels is not a fixed operating configuration. It does not readily permit a falsifiable technical review in the ordinary sense. The question for present purposes is not whether the Applicant could someday supply a more concrete showing. It is whether the Application, as filed, is sufficiently definite to permit ordinary processing now.

B. The Form Instructions and What They Require

The Commission's "Specific Instructions for Schedule S" (SCHS-INST) (OMB Control Number 3060-0678, Edition date August 2024) govern completion of Schedule S in conjunction with FCC Form 312. These instructions are not merely advisory. They are OMB-approved collection requirements under the Paperwork Reduction Act, 44 U.S.C. Section 3507, and they define the structured data the Commission requires to process space-station applications.

The instructions state at Section S3.a that the applicant must provide the "Total Number of Simultaneously Operating Satellites in Constellation," a field auto-populated from the orbital-plane information entered in Section S3.g. Section S3.b requires the "Total Number of Satellites Deployed During the License Term," which must be greater than or equal to S3.a. These are not optional narrative fields. They are structured data entries that drive the Commission's technical review, interference analysis, and staff processing. Here, the Application encodes three satellites in these fields for a system described in the Narrative as up to one million. The Applicant did not request a waiver of the obligation to accurately encode the number of satellites. The Applicant requested a waiver of Schedule S format requirements. Those are not the same thing. The satellite-count fields in Sections S3.a and S3.b are factual inputs, not formatting conventions. Encoding three where one million is described is not a formatting limitation. It is an inaccuracy in a structured data field that the Applicant certified as "true, complete, and correct" under 18 U.S.C. Section 1001.

The same point applies to ITU coordination data. The Applicant's waiver of Schedule S format requirements does not extend to the underlying ITU coordination predicate. The Technical Attachment admits that ITU system information has not yet been submitted. TECH at A-6. The

Applicant did not request a waiver of the ITU filing obligation. That obligation exists independently of Schedule S formatting. It is a filing-stage predicate under the Commission's rules for NGSO applications, and the current record shows it has not been satisfied.

C. The Available Structured Data Methods

The Form Instructions expressly provide two methods for completing Schedule S. The first is manual entry through the ServiceNow software interface. The second is submission through a REST API using JSON formatting. The instructions identify the relevant API functions, including template retrieval, data upload, and read-only JSON retrieval, together with the required authentication and parameter conventions. These are machine-readable structured-data submission methods designed to encode the parameter set of a satellite system in a form the Commission can process.

On the face of the record, either the Applicant did not use the available structured-data method to encode the full system as described, or it did and the encoded data does not match the system described in the Narrative. Either possibility raises the same threshold concern. If OMB-approved structured-data methods existed to encode the system as proposed, the Commission should state why an Application encoding only three satellites was nonetheless docketed and moved to public notice as though the structured record matched the Narrative.

D. The Commission's Own Rules on Return

The Commission's rules are not ambiguous as to the consequence of these conditions. Section 25.112(a) provides that applications not meeting completeness and conformance requirements "shall be returned." The SCHS-INST likewise state: "If information requested on this form is not provided, processing of the application may be delayed or the application may be returned without action pursuant to the Commission rules." The Application contains no ITU coordination data. The Schedule S encodes a system six orders of magnitude smaller than the one described in the Narrative. The Applicant concedes that the instrument "cannot accurately describe" the proposed system. Under the plain language of the Commission's rules and form instructions, this Application should have been returned at the intake stage.

The Commission nonetheless accepted the Application for filing and moved it to public notice within five days. The Objector does not question the Commission's good faith. The Objector asks the Commission to explain, on the record, what limiting principle governs that decision. If an application may be accepted for filing when the applicant concedes that the Commission's standard technical instrument cannot accurately describe the proposed system, when the encoded system is six orders of magnitude smaller than the described system, when ITU coordination data is absent, when the satellite-count fields contain data the Applicant admits is inaccurate, and when multiple waivers displace core filing requirements, then what application could the Commission ever return under Section 25.112?

If the answer is that the Commission exercised discretion to accept the Application notwithstanding these conditions, the Objector respectfully asks the Commission to identify the source and boundaries of that discretion, because neither Section 25.112 nor the Form Instructions appear to provide it where the deficiencies are of the type those provisions were written to address.

The Commission should therefore state whether it finds the Application acceptable for filing under Section 25.112(a), and if so, by what limiting principle. If the Commission instead relies on Section 25.112(b), waiver posture, or some functional equivalent, it should say so expressly and identify the specific basis. The Objector preserves the observation that the Commission's treatment of this Application at the acceptance stage is itself a precedent. If this record is sufficient for acceptance, the Commission has established a new floor, and every future applicant is entitled to know what that floor is.

E. Missing Review Predicates

The missing review predicates identified below are additional to, and not a repetition of, the core Schedule S count and completeness defects discussed above.

Orbital plane information. Section S3.g requires a complete set of orbital-plane parameters for each orbital plane in the constellation, including number of satellites per plane (S3.g.ii), inclination angle (S3.g.iii), orbital period (S3.g.v), apogee and perigee (S3.g.vi through S3.g.ix), and initial phase-angle information for each satellite under Section S3.h. The Application Narrative describes operations across 30-degree and sun-synchronous inclinations at altitudes from 500 to 2,000 km. The Schedule S encodes three satellites in three orbital planes. The orbital-plane data for a million-satellite constellation spanning multiple inclinations and altitude shells does not appear in the structured instrument. The Commission should state whether it considers the orbital-plane information sufficient for technical review and, if not, how it intends to evaluate aggregate effects without it.

Beam-to-orbit mapping. Schedule S Sections S4.u and S5.u require the applicant to provide "a list of each orbital plane in which this antenna beam is used" for both receive and transmit beams. For a constellation spanning multiple altitude shells with potentially different hardware configurations, this mapping is what allows the Commission to associate beam parameters with specific orbital regimes and evaluate interference geometry per shell. The record does not appear to contain a beam-to-orbit association at the scale described. The Commission should state whether beam-to-orbit mapping exists in the record at the actual proposed scale and, if not, how it intends to conduct shell-by-shell interference analysis.

Transmit channel encoding and EIRP. Section S5.x requires channel ID, channel bandwidth, center frequency, and channel type for each transmit channel on each transmit beam. Section S5.j requires maximum transmit EIRP density per beam in the appropriate reference bandwidth. The Application describes one to one hundred dynamic channels with variable bandwidth. A range is not a set of encoded channels. If the channel count, bandwidth, and center frequencies are variable across the described range, the structured fields in S5.x cannot be populated with fixed values that bound the system's emissions behavior. The Commission should state whether the transmit-channel encoding in the filed Schedule S is sufficient to permit aggregate EIRP-density analysis at full constellation scale across the range of described operating configurations.

Receive beam and uplink encoding. Section S4 requires receive-beam parameters including channel bandwidth (S4.x.ii), center frequency (S4.x.iii), and channel type (S4.x.vi) for each receive channel. The Waiver Requests acknowledge that certain uplink-beam channel entries could not be completed due to a form limitation. WAIV at 5. The Commission's own form-validation system requires that "all errors must be corrected before the applicant can electronically file the Form 312 and associated Schedule S." (Form Instructions, Filing Schedule S, Checking for Errors.) The Commission should state whether the acknowledged inability to encode uplink-beam channels was flagged as an error during the validation process, whether the error was overridden, and, if so, under what authority. If the form's own error-checking mechanism identified incomplete uplink entries, acceptance of the filing without resolution of those errors is itself a departure from the Form Instructions that the Commission should explain on the record.

Power-flux-density tables. Section S5.y requires maximum power-flux-density values for each sub-frequency band within each transmit beam, including PFD at specified angles of arrival (S5.y.iv through S5.y.ix) and energy-dispersal bandwidth (S5.y.xiv). These values are what the Commission uses to evaluate compliance with Section 25.208 PFD limits. At one million satellites across multiple orbital shells, the aggregate PFD environment is the sum of individual contributions across the fleet. The Commission should state whether the filed PFD tables reflect the actual proposed system at full constellation density, or whether they reflect only the three-satellite encoding.

Antenna gain contours. Section S8 and the associated beam-attachment requirements (S4.z, S5.z) require predicted antenna-gain contour data for each beam, including GIMS-readable format submissions for interference coordination. For NGSO constellations, the instructions require contours "for one space station if all space stations are identical in the constellation" and a cross-reference file if they are not. The Commission should state whether the filed antenna-gain contour data is representative of the full range of hardware configurations described in the Narrative, and whether the contour data is sufficient for the Commission to evaluate interference at the actual proposed scale rather than at the three-satellite encoding.

The Commission should state on the record whether the technical predicates in the filed Schedule S are sufficient for ordinary review of a system at the scale described in the Narrative. If the

Commission considers the three-satellite encoding a placeholder acceptable under waiver, it should explain how its technical staff intends to conduct interference analysis, PFD compliance review, beam coordination, and aggregate emissions evaluation without structured data at the actual proposed magnitude. The Form Instructions exist because structured data enables review. Where the structured data describes a different system than the one being proposed, the review predicate is absent regardless of what the Narrative says.

F. Environmental and Supervisory Sufficiency

The current record does not resolve whether extraordinary circumstances exist under 47 C.F.R. Section 1.1307(c). Peer-reviewed research documents metallic deposition in the stratosphere from satellite reentry and climate and ozone impacts associated with rocket-launch emissions at constellation scale. Murphy et al., "Metals from Spacecraft Reentry in Stratospheric Aerosol Particles," 120 *Proc. Nat'l Acad. Sci.* e2313374120 (2023); Maloney et al., "The Climate and Ozone Impacts of Black Carbon Emissions From Global Rocket Launches," 127 *J. Geophys. Res. Atmos.* e2021JD036373 (2022). At the scale described in the Application, the environmental question is not abstract. It is whether the Commission can continue to assume ordinary treatment without first stating what categorical exclusion it relies on, whether it has assessed extraordinary circumstances, and whether an Environmental Assessment is required before any grant.

Separate from the environmental issue, the record does not presently appear to contain a concrete showing of deterministic command custody, safe-state behavior, or fail-safe protocols for a crosslink-dependent, autonomous architecture at the scale described. Before the Commission makes a public-interest finding under 47 U.S.C. Section 309(a), it should identify where in the record the operative positive-control showing exists, or say that the present record does not yet contain it.

G. Limiting Principle and Authority

This question is narrow but important. What is the limiting principle and rule-bound source of authority for processing this Application as an ordinary Part 25 matter? The Communications Act authorizes the Commission to regulate communication by wire or radio. 47 U.S.C. Section 153. The Application describes a system whose primary function is artificial intelligence computation, with radiofrequency operations described as backup. NAR at 1 to 2. The Objector does not ask the Commission to decide the entire constitutional perimeter here. The Objector asks only that if the Commission proceeds, it identify the rule-bound source of authority and the limiting principle that contains its action, particularly in light of the Major Questions Doctrine as articulated in *West Virginia v. EPA*, 597 U.S. 697 (2022), and the elimination of Chevron deference in *Loper Bright Enterprises v. Raimondo*, 603 U.S. ___ (2024).

III. FEDERAL ACTIONS AND PROCEEDINGS PLACED ON THE RECORD

The Objector respectfully notes the following federal actions and proceedings, not as fully developed merits arguments, but as federal developments that must be placed on the record during the pleading window because this Application seeks fast-tracked authorization for a system described at civilizational scale and therefore implicates questions of character, qualifications, ownership, deliberative integrity, public interest, national security, treaty-adjacent consequences, and the completeness and reviewability of the administrative record. These matters are included not for distraction, but for preservation. Where an applicant seeks unprecedented authority on an accelerated timetable, record developments bearing on credibility, control, dual-use capability, interagency posture, or federal disclosure context cannot be treated as collateral. They bear directly on whether the Commission may lawfully proceed on the present record, whether additional notice or supplementation is required, and whether the public, including this Objector, has been afforded a meaningful opportunity to test the application actually under review. The Objector raises these matters in his capacity as a citizen, petitioner, and member of the public whose interests in the integrity of the record, the lawful administration of the orbital and spectrum commons, and the qualifications of the entity seeking this authorization are directly implicated by any Commission action in this proceeding.

First, the post-filing ownership and corporate restructuring of the Applicant is placed on the record. On February 2 to 3, 2026, the Applicant's controlling principal completed the acquisition of xAI and X (formerly Twitter) into SpaceX at a combined valuation reported at approximately $1.25 trillion. The controlling principal publicly stated that the merger was "largely about creating space-based data centers." The Ownership Attachment (OWN) filed with this Application was prepared before the merger. The Commission accepted the Application for filing on February 4, 2026 on the basis of a pre-merger ownership record. The Objector places this development on the record because it bears directly on the adequacy of the Ownership Attachment, the completeness of the administrative record, the current ownership and control picture, the foreign-investment posture of the entity actually seeking authorization, and whether additional notice or supplementation is required before the Commission acts. In a fast-tracked proceeding seeking authorization at this magnitude, changes to applicant identity, control, capital structure, and related disclosure posture are not peripheral. They are necessary preservation matters because they go to who the Commission is being asked to trust, authorize, and supervise.

Second, during the period relevant to this Application and the surrounding federal events, the Applicant's controlling principal had recently served, or was publicly identified as serving, as a Special Government Employee in connection with DOGE. Public reporting and litigation concerning DOGE personnel and DOGE-related access or ethics questions at federal agencies, including the FCC, are placed on the record only insofar as those matters may bear on deliberative integrity, conflict screening, consultation history, recusals, prioritization, or the public's ability to understand who participated in agency review. These matters are necessary to preserve because a proceeding of this scale and strategic significance cannot rest on an opaque decisional path. Where the applicant's controlling principal has been publicly linked to temporary federal service and related ethics controversies, the public and the Objector alike have a legitimate interest in whether the review process was insulated, disclosed, and conducted with procedural integrity.

Third, the February 27, 2026 Defense Department designation of Anthropic as a supply-chain risk to national security is placed on the record to the extent this Application is framed around orbital infrastructure for advanced artificial intelligence and large-scale space-based computing. Public reporting indicates that the dispute involved restrictions concerning autonomous weapons and related military uses, including space-based defense concepts. The Objector does not ask the Commission to adjudicate the merits of that separate federal action here. The Objector asks only that the Commission not proceed as though no such federal action exists when evaluating an application to place unprecedented, potentially dual-use AI computing infrastructure in orbit, where questions of public interest, national-security externalities, dual-use capability, strategic dependence, and treaty-adjacent consequences may overlap. This matter is necessary to preserve because the Commission is being asked to authorize not an ordinary communications system, but infrastructure that the Applicant itself describes in terms that touch advanced AI, orbital computing, and civilizational ambition. At that scale, the national-interest dimension is not collateral to the public-interest inquiry.

Fourth, documents released during the comment period pursuant to the Epstein Files Transparency Act of 2025 are placed on the record to the extent they may bear on the character qualification inquiry under 47 U.S.C. Section 308. Bloomberg reported on February 23, 2026 that released documents include records concerning the Applicant's controlling principal and SpaceX's ITAR-controlled facility. The Objector makes no characterization of any person. The Objector observes that Section 308 requires applicants to demonstrate the qualifications the public interest requires, and that the character-qualification record may be materially supplemented by federal records during the Commission's deliberation period. The Commission should state whether it has considered these records, or state that it has not. This matter is necessary to preserve because the Commission cannot meaningfully separate a fast-tracked request for unprecedented orbital authority from the qualifications and credibility of the persons and entities behind it. Where potentially relevant federal records emerge during the pleading window, omission of those developments would leave the record artificially incomplete on a subject the Communications Act itself treats as material.

Fifth, the federal UAP disclosure and records framework reflected in the February 19, 2026 Presidential directive and related NDAA provisions (FY2022 through FY2026), including the establishment of AARO (50 U.S.C. Section 3373), the NARA UAP Records Collection (NDAA FY2024 Sections 1841 through 1843), and UAP-specific whistleblower protections that expressly nullify non-disclosure agreements (NDAA FY2024 Section 1673), is placed on the record. The Objector does not ask the Commission to adjudicate that framework in this filing. The Objector asks only that the Commission recognize that it exists as part of the broader federal backdrop in which this Application is being reviewed. This matter is preserved because the Application concerns orbital infrastructure at extraordinary scale, and the federal government's own disclosure, records, and whistleblower frameworks may bear on what information can lawfully remain undisclosed, what reporting barriers are ineffective as a matter of law, and what broader national-interest context surrounds the Commission's consideration of activities in orbital space.

Sixth, the COVID-19 declassification framework and a documented COVID-19-era discrepancy concerning public technical representations by the Applicant's controlling principal are placed on the record for preservation, credibility, and self-certification purposes. Under NDAA FY2026, P.L. 119-60, Section 6803, the Director of National Intelligence and relevant intelligence-community elements must conduct a declassification review concerning the origins of COVID-19 within 180 days of enactment, meaning additional government records of potential relevance may surface during or near the Commission's deliberation period. Separately, public reporting in April 2020 documented that devices publicly described as "ventilators" were, in multiple reported instances, BiPAP or CPAP breathing-support machines rather than full ventilators. The Objector does not ask the Commission to adjudicate either matter here as a standalone issue. The point for present purposes is narrower and preservational: this proceeding likewise depends heavily on applicant self-description, self-certification, and the Commission's willingness to rely on representations about what the system is, what it can do, and what supporting safeguards exist. In a fast-tracked application seeking unprecedented authority at civilizational scale, both convergent federal disclosures and documented prior instances in which public technical representations diverged from what recipients or reviewers understood them to mean are properly preserved as part of the broader qualifications, credibility, and public-interest context. The matter is therefore placed on the record to the extent it may bear on the reliability of present representations, the weight the Commission should place on unsupported assurances, and whether additional federal records surfacing during the Commission's review should be considered before action is taken.

IV. RECORD OMISSIONS BEARING ON THE PUBLIC INTEREST DETERMINATION

The Objector respectfully identifies the following omissions from the current record. These are observations that the record, as filed, does not appear to contain certain baseline inputs the Commission would ordinarily need in order to make the public-interest finding required by 47 U.S.C. Section 309(a). For each, the Objector asks a simple question: how does the Commission intend to weigh the record without it?

System classification. The record does not clearly state whether the proposed architecture is a standalone payload, a modification of the existing Starlink constellation, or a hybrid architecture requiring integrated review. This is not a peripheral matter. It affects what rules apply, what coordination is required, what environmental-review posture governs, and whether the Commission is reviewing a discrete system or part of a larger connected action. The Application describes a system whose primary data pathway is optical inter-satellite links routed through the separately licensed Starlink constellation. NAR at 2; TECH at 1 to 2. If the proposed system cannot operate independently of Starlink, the Commission should explain why it is being reviewed as an independent application rather than as a modification requiring integrated analysis of both systems.

Ground segment authorization and command-path independence. The Application's operational architecture depends on earth stations and command pathways that do not appear to be independently authorized for this system. The Technical Attachment describes a Ka-band TT&C backup pathway that would communicate directly with SpaceX's authorized Gen1 and Gen2 Ka-band earth stations. TECH at A-2 to A-3. Those earth stations are authorized under a separate license for a different system. The record does not contain a showing that cross-system use is permitted under the existing authorizations, nor does it identify a dedicated, independent ground segment for telemetry, tracking, and command that does not depend on separately licensed infrastructure. The Application also describes a system that relies on the separately licensed Starlink constellation for its primary data pathway, including optical inter-satellite links routed through Starlink infrastructure. NAR at 2; TECH at 1 to 2. The Commission should state whether it considers the ground-segment and command-path showing sufficient and, if so, under what authority cross-system reliance satisfies the points-of-communication requirement and how positive control and independent command custody would be ensured for one million satellites if the command pathway depends on infrastructure authorized under a different license with different terms and conditions.

Post-merger ownership. The Ownership Attachment was prepared and filed before the February 2 to 3, 2026 SpaceX/xAI merger. The entity described in the Ownership Attachment is not the entity that exists today. The Commission's acceptance-for-filing determination was made on the basis of an ownership representation that is now materially stale. The Commission should state whether it considers a pre-merger Ownership Attachment sufficient for a post-merger applicant and, if so, why renewed notice is not required to allow the public to evaluate the current ownership, control, and foreign-investment positions of the entity actually seeking authorization.

Form 312 certification. The signatory certified on Form 312, under 18 U.S.C. Section 1001, that the Application is "true, complete, and correct." The same certification covers the environmental statement, which asserts no significant environmental impact for a system requiring approximately 200,000 atmospheric reentries per year at steady state. Published peer-reviewed research contradicts that assertion at the lifecycle throughput described. The Commission should state whether it considers a certification "true, complete, and correct" when the Applicant simultaneously concedes that the Commission's standard technical instrument cannot accurately describe the proposed system, when the encoded satellite count is six orders of magnitude below the described system, and when the environmental certification is contradicted by published science. If the Commission accepts the certification notwithstanding these conditions, it should explain what standard governs the "true, complete, and correct" requirement going forward.

Waiver stack as structural posture. The Application requests waivers of multiple Part 25 requirements simultaneously, including Sections 25.156, 25.157, 25.164, 25.165, and 25.114. WAIV. Four of the five waivers address core structural filing requirements rather than routine operational accommodations. That does not automatically defeat the filing. But where the cumulative effect of the waiver stack is to displace the structured data inputs that make an authorization evaluable, enforceable, and reviewable, the question arises whether the standard Part 25 form and ordinary processing path are the right instrument for what is actually being proposed. The Commission should state on the record whether it considers the present waiver architecture sufficient for ordinary review and, if so, on what specific basis, including an explanation of how staff intends to draft enforceable license conditions against parameters that have been waived out of the required disclosure instrument.

Launch cadence, lifecycle throughput, and atmospheric loading. The Application describes a system of up to one million satellites with a five-year design lifetime, implying a steady-state replacement rate of approximately 200,000 satellites per year. The record does not appear to contain a bounded lifecycle-throughput envelope stating the annualized deployment cadence, the replacement rate, the disposal and atmospheric-reentry throughput, or the aggregate reentry mass per year. How does the Commission intend to evaluate cumulative environmental, spectrum, and orbital-commons impacts without these inputs? Recent peer-reviewed research has now demonstrated that atmospheric contamination from a single rocket-stage reentry is directly measurable. Wing et al. detected a tenfold enhancement of lithium atoms at 96 km altitude following the uncontrolled reentry of a single Falcon 9 upper stage, representing the first time-resolved measurement of upper-atmospheric pollution from space-debris reentry. Wing et al., "Measurement of a Lithium Plume from the Uncontrolled Re-entry of a Falcon 9 Rocket," 7 Communications Earth & Environment 161 (2026). At the Application's described steady-state replacement rate, this measurable contamination pathway would recur approximately 200,000 times per year. The Commission should state how it intends to evaluate cumulative atmospheric loading from reentry at this cadence without a bounded lifecycle-throughput envelope in the record.

AI governance and autonomous-operation safeguards. The Application proposes to provide "unprecedented computing capacity to power advanced artificial intelligence models." NAR at 1. The record does not appear to contain any AI-governance framework, any description of autonomous-operation constraints, or any mechanism by which the Commission, the Applicant, or any other body could intervene to halt, restrict, or redirect AI operations conducted from orbital infrastructure if circumstances required it. How does the Commission intend to make a public-interest finding for an orbital AI platform without identifying what governance or intervention framework applies?

Oversight and enforcement capacity at scale. The Commission's existing oversight infrastructure was designed for communications satellite constellations, not for a system of one million autonomous computing platforms. The record does not address what oversight resources would be required, what enforcement mechanisms would apply at this scale, or whether the Commission's current staffing and technical capacity are adequate to supervise the authorization as described. How does the Commission intend to ensure that a grant at this magnitude does not exceed its practical capacity to enforce compliance?

Space-weather resilience and degraded-state behavior. The record does not appear to contain a severe space-weather scenario, a correlated-anomaly tolerance, or a degraded-state emissions model describing system behavior when solar events, geomagnetic storms, or other space-weather conditions affect fleet operations. For a system of this density, the Commission cannot bound aggregate interference or emissions behavior under stress without these inputs. How does the Commission intend to evaluate worst-case behavior without a worst-case envelope?

Fast Radio Burst and passive-band protection at fleet scale. Fast Radio Bursts are non-repeatable transient astrophysical events lasting milliseconds. They are radio-frequency phenomena whose detection depends on low aggregate noise floors. 47 C.F.R. Section 2.106, footnote 5.340, prohibits all emissions in passive bands. The record does not appear to bound out-of-band or spurious emissions at fleet scale across the range of dynamic channel configurations described. A transient event masked by a satellite's emission during that millisecond window is permanently lost. How does the Commission intend to satisfy its passive-band protection obligations without a bounded fleet-scale emissions envelope?

The Objector observes that the Applicant possesses advanced engineering and computational resources sufficient to produce technical data in a format the Commission and the public can evaluate. Does the Commission consider that an applicant proposing a system of this complexity bears the burden at the time of filing to produce data adequate for the Commission's review, rather than to defer that production to post-grant conditions or supplemental filings? Does the Commission consider the public interest better served when proposed systems are engineered to satisfy applicable legal and environmental requirements at the time of filing, rather than to seek waiver of those requirements? Does the Commission agree that the scope and rigor of an applicant's technical showing should be commensurate with the scale and complexity of the system proposed?

V. APPROPRIATIONS AND FISCAL PREDICATE QUESTIONS

The Objector respectfully raises the following questions concerning the fiscal and administrative implications of a grant at this scale.

If the Commission grants this Application, the authorization may require substantial federal resources for supervision, enforcement, spectrum monitoring, interagency coordination, and any additional review or compliance activity the Commission determines is necessary. The Objector does not allege that any violation has occurred. The Objector asks only whether the Commission has determined that the oversight and supervisory demands reasonably foreseeable at one-million-satellite scale can be carried out within existing lawful authority, existing appropriations, and existing administrative capacity.

The Anti-Deficiency Act provides that federal officers and employees may not make or authorize expenditures or obligations in excess of available appropriations, may not involve the Government in obligations to pay money before appropriations are made unless authorized by law, and may not accept unauthorized voluntary services. 31 U.S.C. Sections 1341, 1342. The Objector therefore asks whether the Commission has considered the fiscal predicate for any supervisory, monitoring, enforcement, interagency, or compliance burden that would foreseeably accompany an authorization of this magnitude, and whether any such burden has been assessed as compatible with currently available appropriations and lawful authority.

A conditional grant that leaves essential oversight architecture, supervisory mechanisms, or resource commitments to be developed later may present not only a regulatory gap, but a fiscal and administrative one. The Commission should therefore state on the record what fiscal or resource analysis, if any, it has conducted regarding the practical costs of supervising and enforcing compliance for an authorization at this scale.

To the extent any DOGE-affiliated personnel, detailees, contractors, or interagency resources materially participated in review or handling of this Application, or exercised influence over rulemaking bearing on the processing of applications of this type, the Objector respectfully asks whether the Commission has determined that any such participation rested on an adequate legal and fiscal predicate, including any necessary detail authority, reimbursement authority, appropriation, or other lawful basis.

VI. AUDIT, SELF-CERTIFICATION, AND THE STARLINER PRECEDENT

The Objector respectfully raises the following questions concerning the Commission's reliance on self-certification for a system of this scale and complexity.

On February 19, 2026, NASA released its final report on the Starliner Crewed Flight Test Investigation. The Starliner findings are not directly about this Applicant or this system. They are relevant here only in a narrower sense: they show that in complex orbital programs, internal certification and quality-assurance processes can fail in ways that are not fully visible on the face of the record until the risks become operationally significant. NASA stated that the investigation identified an interplay of hardware failures, qualification gaps, leadership missteps, and cultural breakdowns in a mission-critical space system operated by a major aerospace contractor.

This Application likewise relies heavily on self-certification. The Form 312 certification states that the Application is "true, complete, and correct" under 18 U.S.C. Section 1001. The environmental certification on Form 312 states that a system requiring approximately 200,000 atmospheric reentries per year at steady state will have no significant environmental impact. The technical certifications likewise rest on the Applicant's own representations regarding command custody, spectrum compliance, and orbital-debris mitigation.

The Objector does not ask the Commission to reject self-certification as a regulatory tool. The Objector asks the Commission to explain, on the record, what independent audit, verification, or validation framework, if any, it intends to require for a system of this scale before or after grant. Does the Commission intend to rely solely on the Applicant's self-reporting for orbital-debris compliance, spectrum compliance, environmental compliance, and operational control at one-million-satellite scale? If so, has the Commission considered whether that level of reliance is adequate for an unprecedented orbital system? If the Commission intends to require independent verification, where in the record is the framework for that verification described?

The Objector observes that the Applicant has the engineering capability to satisfy reasonable audit or verification requirements. The question is not capability. The question is whether the Commission's current record contains the framework, or whether the Commission intends to grant first and develop the framework later. If the latter, the Commission should say so on the record and explain why that sequence is consistent with the public-interest standard of Section 309(a). Filing-stage predicates identified in this objection cannot be converted into post-grant conditions where the missing predicates are themselves part of what the Commission would need in order to determine that grant serves the public interest in the first instance.

VII. ADDITIONAL RECORD-CLARIFICATION QUESTIONS

The Objector respectfully raises the following limited questions for record clarity. They are stated because they have been publicly reported or formally litigated elsewhere and may bear on applicant credibility, interagency coordination, public-interest review, or the completeness of the Commission's own deliberative record.

First, to what extent, if at all, has the Commission considered public reporting and independent radio-frequency observations indicating that satellites associated with the separately authorized

Starshield constellation may be transmitting space-to-Earth in the 2025 to 2110 MHz band, a use described as potentially inconsistent with ordinary ITU spectrum allocations and potentially relevant to interference risk if confirmed? Geoff Brumfiel, "A Classified Network of SpaceX Satellites Is Emitting a Mysterious Signal," NPR (Oct. 17, 2025); L. Scott Tilley, "Detection of Strong S-Band Emissions from the Starshield Constellation: Observations and Regulatory Context," Zenodo, Record 17373141 (Oct. 17, 2025). If the Commission has considered those reports, it should say so on the record. If it has not, it should say that instead.

Second, to the extent any aspect of the Applicant's operations, including operations associated with the separately authorized Starshield constellation, involves classified programs or national-security designations, the Objector respectfully asks the Commission to clarify whether any such classification posture affects its evaluation of compliance with the Commission's rules, the Communications Act, or applicable international spectrum allocations. Executive Order 13526, Section 1.7(a)(1), provides that information may not be classified in order to conceal violations of law, inefficiency, or administrative error. The Objector raises this point only to preserve the principle that classification status does not itself resolve, excuse, or displace otherwise applicable regulatory obligations.

Third, the Objector respectfully notes that NDAA FY2024 Section 1673 (50 U.S.C. Section 3373 note) expressly nullifies any non-disclosure agreement, order, or other instrument that would prohibit or restrict reporting to Congress, Inspectors General, AARO, or other authorized recipients of information relevant to the matters raised in this proceeding. To the extent any participant in this proceeding, including Commission staff, the Applicant, or its contractors, possesses information bearing on the questions raised herein and believes that a non-disclosure agreement prevents its disclosure to an authorized recipient, that belief would be inconsistent with the federal disclosure framework the Objector has placed on the record.

Fourth, to what extent, if at all, has the Commission considered conflict-screening, recusal, consultation-history, or deliberative-integrity questions associated with DOGE personnel or DOGE-related access and ethics controversies at the FCC, insofar as those questions could bear on consultation, prioritization, recusals, or process integrity in a proceeding involving a Musk-controlled regulated entity? During the period relevant to this Application and the surrounding federal events, the Applicant's controlling principal had recently served, or was publicly identified as serving, as a Special Government Employee in connection with DOGE. The Objector raises this point only insofar as it may bear on conflict-screening, consultation history, recusals, or deliberative integrity during the period relevant to this Application.

VIII. Alternative Procedural Relief: Extension, Reopening, or Supplemental Notice

The Objector respectfully requests that the Commission extend or reopen the pleading period, or otherwise require supplementation and renewed public notice to the extent the Commission concludes that material post-filing developments altered the posture the public was asked to evaluate. The public notice process in Part 25 exists to permit meaningful public testing of the application as noticed, and the Commission's rules expressly contemplate both public notice and, where appropriate, extension of the filing deadline.

Material developments bearing on the Commission's public-interest review arose during the original thirty-day window. These include asserted post-filing changes affecting ownership or control, as well as other developments the Objector contends bear directly on the adequacy of the record and the Commission's ability to evaluate the application as noticed. On that basis, the Objector respectfully requests a reopened or extended opportunity for public comment, or other procedural relief sufficient to ensure that the record reflects the application the Commission is actually being asked to act upon.

If the Commission declines this procedural relief, the Objector respectfully requests that the Commission so state in any order disposing of the application and explain whether it concluded that no material supplementation, renewed notice, or additional public process was warranted.

IX. Alternative Merits Relief if Procedural Relief Is Denied or Deferred

If the Commission does not extend or reopen the pleading period, and does not otherwise require procedural measures sufficient to cure the present defects in the record, the Objector respectfully requests the following relief in the alternative.

First, the Commission should return the application as unacceptable for filing under 47 C.F.R. § 25.112(a), or otherwise dismiss it without prejudice, if it concludes the current record is incomplete, internally inconsistent, or not sufficiently evaluable for ordinary processing.

Second, if the Commission does not return or dismiss the application, it should hold the matter in abeyance and require supplementation, clarification, or amendment sufficient to resolve the threshold defects identified in this filing, including any supplementation the Commission finds necessary to reflect post-filing developments affecting the application as noticed.

Third, if substantial and material factual questions remain, or if the Commission cannot make the findings required by 47 U.S.C. § 309(a), it should designate the matter for hearing under 47 U.S.C. § 309(e).

Fourth, whatever disposition the Commission reaches, the Objector respectfully requests express findings on the threshold questions raised in this filing, including the procedural path the Commission used to reach its decision.

X. PRESERVATION, ANTI-SUPPLEMENTATION, AND RECORD CLARIFICATION

This filing is reduced for clarity, not waiver. The Objector preserves arguments not fully developed here to the extent they become relevant in any later administrative or judicial proceeding.

Any post-window submission by the Applicant that supplies information identified herein as missing, incomplete, or structurally deficient, and that materially changes the interference analysis, operative system description, enforceable license conditions, ownership or control picture, or the Commission's ability to supervise the proposed system, should be treated as a material amendment and processed accordingly. If the Commission permits supplementation during or after the pleading period, commenting parties should be afforded a meaningful opportunity to evaluate and respond to the operative record, including extension or reopening of the pleading cycle if necessary.

To the extent this proceeding is subject to the Commission's permit-but-disclose ex parte rules, the Objector respectfully requests that any ex parte presentations concerning the substance, timing, scope, or treatment of supplemental filings be disclosed in the record in accordance with the Commission's rules.

Respectfully submitted,

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APPENDIX A: NOTICE OF PRESERVED MATTERS

The following matters have been identified in the Objector's research and are preserved for development in any future proceeding. This list is organized by subject matter. The gate-keyed analytical architecture underlying these matters is summarized in Appendix C.

1. Schedule S magnitude mismatch and filing-instrument capacity.
2. ITU coordination posture and missing international review predicates.
3. Taxonomy and classification of the proposed system, including its relationship to Starlink and any connected-action analysis.
4. Environmental review at scale, including cumulative atmospheric loading, reentry deposition, rocket emissions, and any need for Environmental Assessment or broader NEPA review.
5. Positive control, command custody, fail-safe behavior, and administrability of post-grant supervision.
6. Limiting principle, statutory authority, and major-questions concerns raised by a compute-dominant orbital architecture.
7. Post-filing ownership restructuring, applicant identity, and the adequacy of the pre-merger Ownership Attachment, including any bearing on IPO-related disclosure adequacy.
8. Character qualification under 47 U.S.C. Section 308, including any bearing of documents released during the comment period.
9. Launch cadence, lifecycle throughput, and the absence of a bounded reentry and disposal envelope.
10. AI governance, autonomous-operation safeguards, and the absence of any intervention or constraint framework for orbital AI operations.
11. Oversight capacity, enforcement resources, and the fiscal predicate for supervision at the described scale.
12. Space weather resilience, degraded-state behavior, and the absence of a worst-case emissions envelope.
13. Fast Radio Burst protection and passive-band compliance at fleet scale.
14. Dedicated ground segment, command-path independence, and the dependency on separately licensed Starlink infrastructure.
15. Audit, independent verification, and the adequacy of self-certification in light of the Starliner Crewed Flight Test Investigation findings.
16. The Anti-Deficiency Act dimension: whether a grant at this scale foreseeably creates obligations exceeding the Commission's existing appropriations.
17. Sovereignty and strategic control concentration, including treaty-linked liability under the Outer Space Treaty Articles VI and VII, supply-chain dependency, orbital enclosure risk, and cascade-failure exposure arising from single-entity authorization at this scale.
18. The adequacy of the Commission's electronic filing and docketing procedures, including whether available structured-data submission methods should have been required when the standard Schedule S instrument was acknowledged as inaccurate.
19. The implications of the changing corporate, ownership, and merger securities environment surrounding the Applicant
20. All 8 Threshold Screens, 6 Acceptability Checks, 7 Record Sufficiency Gaps, and 5 Fatal Flaws organized across the Objector's six-gate analytical architecture, as summarized in Appendix C.

21. All 30 Discrepancy Forks (DC-1 through DC-30) identified in the Objector's research and summarized in Appendix C.
22. All 9 Public Interest Forks (PI-1 through PI-9) identified in the Objector's research and summarized in Appendix C.

Legal Authorities (Principal Cases)

Motor Vehicle Mfrs. Ass'n v. State Farm Mut. Auto. Ins. Co., 463 U.S. 29, 43 (1983). SEC v. Chenery Corp., 332 U.S. 194 (1947). West Virginia v. EPA, 597 U.S. 697 (2022). Loper Bright Enterprises v. Raimondo, 603 U.S. ___ (2024). Learning Resources, Inc. v. Trump, 607 U.S. ___ (2026). WAIT Radio v. FCC, 418 F.2d 1153 (D.C. Cir. 1969). Delaware Riverkeeper Network v. FERC, 753 F.3d 1304 (D.C. Cir. 2014). Policy Regarding Character Qualifications in Broadcast Licensing, 102 FCC 2d 1179 (1986).

Preservation of Constitutional, Treaty, International, Technical, Scientific, and AI-Related Arguments.

The Objector preserves all arguments arising under the Constitution of the United States, including the First Amendment, the Due Process Clause, the Non-Delegation Doctrine, and the Treaty Clause. The Objector preserves all arguments arising under international treaties and agreements to which the United States is a party, including the Outer Space Treaty (1967), the Liability Convention (1972), the Registration Convention (1975), the ITU Constitution, Convention, and Radio Regulations, and the United Nations Guidelines for the Long-term Sustainability of Outer Space Activities (2019). The Objector preserves all arguments concerning the need for independent technical and scientific expert review of the Application's system architecture, interference methodology, orbital debris modeling, atmospheric reentry impacts, passive-band protection, positive-control architecture, and AI governance framework, including review by the National Academies, NASA, NOAA, the National Science Foundation, or other qualified independent bodies as appropriate or warranted. The implications of the changing corporate, ownership, and pre-IPO securities environment surrounding the Applicant are preserved for additional findings in any appropriate forum. No argument is waived by its non-development in this filing.

APPENDIX B: SELECTED STUDIES AND REPORTS CITED OR RESERVED

Atmospheric and Environmental

Murphy et al., "Metals from Spacecraft Reentry in Stratospheric Aerosol Particles," 120 Proc. Nat'l Acad. Sci. e2313374120 (2023). Detection of spacecraft-origin metals in stratospheric aerosol particles at concentrations above natural background.

Maloney et al., "The Climate and Ozone Impacts of Black Carbon Emissions From Global Rocket Launches," 127 J. Geophys. Res. Atmos. e2021JD036373 (2022).

Maloney et al., "Investigating the Potential Atmospheric Accumulation and Radiative Impact of the Coming Increase in Satellite Reentry Frequency," 130 J. Geophys. Res. Atmos. e2024JD042442 (2025). Models accumulation pathway and radiative forcing from megaconstellation reentry alumina at projected scale.

Wing et al., "Measurement of a Lithium Plume from the Uncontrolled Re-entry of a Falcon 9 Rocket," 7 Comms. Earth & Env't 161 (2026). First ground-based lidar detection of upper-atmospheric pollution from a specific rocket stage reentry.

Jain, Eastham, and Hastings, "Future of Satellite Reentry and Earth's Atmosphere: the Lifetime and Direct Radiative Forcing of Space Debris Reentry Alumina," ESS Open Archive (2023), doi:10.22541/essoar.170110669.90121203/v1.

"Determining Mass Fluxes of Space Debris upon Demise in the Atmosphere," Acta Astronautica (2025), doi:10.1016/j.actaastro.2025.09.036.

Space Policy and Systems

NASA, Program Investigation Team Report on the Boeing CST-100 Starliner Crewed Flight Test (released Feb. 19, 2026). Reserved as analogous safety-governance authority regarding hardware failures, qualification gaps, and cultural breakdowns in mission-critical space programs.

Kardashev, N.S., "Transmission of Information by Extraterrestrial Civilizations," 8 Soviet Astronomy 217 (1964). Reserved as the original scientific framework invoked by the Applicant in characterizing the proposed system's scale.

Spectrum, Orbital Tracking, and Starshield

Tilley, L.S., "Detection of Strong S-Band Emissions from the Starshield Constellation: Observations and Regulatory Context," Zenodo Record 17373141 (Oct. 17, 2025).

Brumfiel, G., "A Classified Network of SpaceX Satellites Is Emitting a Mysterious Signal," NPR (Oct. 17, 2025).

Clark, S., "Pentagon May Put SpaceX at the Center of a Sensor-to-Shooter Targeting Network," Ars Technica (July 1, 2025).

Post-Filing Federal Developments and Character Qualification

Hull, D., "Epstein Sought to Bring Entourage of Women on 2013 SpaceX Visit," Bloomberg (Feb. 23, 2026). DOJ-released records bearing on character qualification under 47 U.S.C. Section 308 and ITAR compliance history.

CNN (Jan. 25, 2026); PBS/AP (Jan. 26, 2026); Al Jazeera (Jan. 26, 2026). Reporting on classified directed-energy weapon deployment and AI-weapons convergence.

CNN (Apr. 17, 2020); The Hill (Apr. 17, 2020); Washington Post (Apr. 29, 2020). Reporting on ventilator/BiPAP discrepancy bearing on self-certification reliability.

CNBC (Feb. 3, 2026). Post-filing SpaceX/xAI merger at reported $1.25 trillion valuation; controlling principal's statement that the merger is "largely about creating space-based data centers."

Bloomberg (Feb. 3, 2026; Dec. 9, 2025; Mar. 3, 2026). SpaceX/xAI merger and IPO reporting.

CNBC (Feb. 11, 2026). xAI regulatory probes bearing on character qualification.

DOGE, SGE, and Institutional Integrity

Burleigh v. FCC (D.D.C., filed Apr. 2025; order Aug. 27, 2025). Court found FCC DOGE-related record production "vague and uninformative."

Communications Daily (Aug. 27, 2025); NBC News (July 25, 2025); AFL-CIO Report (June 24, 2025); American Prospect (July 14, 2025). Reporting on DOGE personnel conflicts and SpaceX board member service in executive branch.

AI Governance

CBS News (Feb. 27, 2026); Fortune (Feb. 28, 2026); Just Security (Mar. 2, 2026); Lawfare (Mar. 2, 2026). Reporting and legal analysis on Anthropic supply-chain risk designation and AI governance vacuum.

APPENDIX C: SIX-GATE ANALYTICAL ARCHITECTURE PRESERVED

The Objector's research is organized across a six-gate analytical framework totaling 65 independently sufficient bases for Commission action. Each gate corresponds to a stage in the Commission's own review sequence. The full substance of each gate is preserved and available for submission in any future proceeding. Nothing in this appendix constitutes waiver of any matter identified herein.

Gate 1: Threshold Screens (TS-1 through TS-8)

Eight antecedent questions the Commission must resolve affirmatively before processing this Application. Failure of any screen means the Commission lacks a prerequisite to proceed.

TS-1. Classification and limiting principle. 47 C.F.R. Sections 25.112, 25.114; 47 U.S.C. Sections 151, 309(a).
TS-2. Major Questions Doctrine. 47 U.S.C. Section 309(a); 5 U.S.C. Section 706.
TS-3. ITU coordination and international posture. 47 C.F.R. Sections 25.111(b), 25.114.
TS-4. Bounded parameters and reviewability. 47 C.F.R. Section 25.114; 5 U.S.C. Section 706.
TS-5. Passive and prohibited allocation protection. 47 C.F.R. Section 2.106, fn. 5.340; 47 C.F.R. Section 25.114.
TS-6. Command custody and positive control. 47 C.F.R. Sections 1.17, 25.112.
TS-7. NEPA compliance posture. 42 U.S.C. Section 4332; 47 C.F.R. Section 1.1307.
TS-8. Independent verification and audit. 47 C.F.R. Section 1.17; 47 U.S.C. Sections 308(b), 309(d).

Gate 2: Acceptability Checks (AC-1 through AC-6)

Six independently sufficient facial defects requiring return or dismissal under 47 C.F.R. Section 25.112(a).

AC-1. Schedule S encodes three satellites for a noticed million-satellite system.
AC-2. The Applicant concedes the filing instrument cannot accurately describe the system.
AC-3. No beam and channel plan encoded with sufficient specificity for falsifiable interference analysis.
AC-4. ITU coordination information absent at filing.
AC-5. Ground segment and points-of-communication authority unresolved.
AC-6. Waiver stack displaces the structured disclosures that make public review possible.

Gate 3: Record Sufficiency Gaps (RS-1 through RS-7)

Seven baseline inputs the Commission needs before making a Section 309(a) finding. These are filing-stage predicates, not post-grant deliverables.

RS-1, worst-case operating envelope including space-weather scenarios;
RS-2, fleet-wide aggregate RF occupancy baseline;
RS-3, out-of-band and spurious emissions envelope with passive-band protection;

RS-4, replacement-continuity for displaced sensing functions and non-replayable observation windows;
RS-5, conjunction screening and correlated-failure analysis at scale;
RS-6, bounded lifecycle cadence, disposal throughput, and reentry mass envelope;
RS-7, independent verification and audit regime.

Governing authorities: 47 C.F.R. Sections 25.114, 25.114(d)(14), 2.106 fn. 5.340, 1.17; 47 U.S.C. Section 303(f); ITU-R RA.769; NEPA; 5 U.S.C. Section 706.

Gate 4: Fatal Flaws (FF-1 through FF-5)

Five structural legal bars arising at the level of agency authority, statutory classification, and constitutional structure. These are not record gaps curable by supplementation, conditions, or post-grant amendment. Each is independently sufficient.

FF-1. Forcing the Major Questions Doctrine. West Virginia v. EPA, 597 U.S. 697 (2022); Biden v. Nebraska, 600 U.S. 477 (2023); Loper Bright Enterprises v. Raimondo, 603 U.S. ___ (2024); Consumers' Research v. FCC, 45 F.4th 816 (5th Cir. 2022); Learning Resources, Inc. v. Trump, 607 U.S. ___ (2026); State Farm, 463 U.S. at 43; 47 U.S.C. Sections 303(f), 309(a); 5 U.S.C. Section 706.

FF-2. Authority Collapse of Constitutional Jurisdiction. Outer Space Treaty Articles II, VI, VII, IX; Gundy v. United States, 139 S. Ct. 2116 (2019); Carpenter v. United States, 585 U.S. 296 (2018); United States v. Jones, 565 U.S. 400 (2012); Kyllo v. United States, 533 U.S. 27 (2001); 47 U.S.C. Section 303(f); 5 U.S.C. Section 706.

FF-3. Taxonomic Failure Through Categorical Misclassification. 47 U.S.C. Sections 151, 153; 47 C.F.R. Section 25.112; State Farm, 463 U.S. at 43; 5 U.S.C. Section 706.

FF-4. Architectural Incapability of Licensure of Open-Loop Systems. State Farm, 463 U.S. at 43; Delaware Riverkeeper Network v. FERC, 753 F.3d 1304 (D.C. Cir. 2014); 47 C.F.R. Section 25.114; 42 U.S.C. Section 4332 (NEPA); 5 U.S.C. Section 706.

FF-5. Loss of Sovereignty Through Unlawful Delegation of Strategic Control. Youngstown Sheet and Tube Co. v. Sawyer, 343 U.S. 579 (1952); Outer Space Treaty Articles VI, VII; State Farm, 463 U.S. at 43; 15 C.F.R. Sections 734.13, 734.15; 22 C.F.R. Section 120.17; 47 U.S.C. Section 303(f); 5 U.S.C. Section 706.

Full analytical development for each flaw is preserved for any future proceeding.

Gate 5: Discrepancy Forks (DC-1 through DC-30)

Thirty decision-branch questions arising from internal contradictions within the Applicant's own filings. Each presents a binary choice: act on the existing record or require material cure triggering renewed notice and a reset pleading cycle.

Cluster 1: Schedule S and Encoding Evaluability (DC-1 through DC-5).
Cluster 2: ITU and International Coordination (DC-6 through DC-10).
Cluster 3: Command, Control, and Auditability (DC-11 through DC-15).
Cluster 4: Safety, Debris, and Lifecycle (DC-16 through DC-20).
Cluster 5: Environmental and Alternatives (DC-21 through DC-25).
Cluster 6: Notice Integrity and Timing (DC-26 through DC-30), including the DC-30 Timing Guardrail.

Governing authorities: 47 C.F.R. Sections 25.112, 25.114, 25.116, 25.283, 1.17, 1.1307, 2.106; 47 U.S.C. Sections 309(d), 303(f); NEPA; Outer Space Treaty; 5 U.S.C. Section 706. Full development preserved.

Gate 6: Public Interest Forks (PI-1 through PI-9)

Nine public-interest findings the Commission must make before any grant under 47 U.S.C. Section 309(a), none of which can be made on the present record: PI-1, orbital commons enclosure; PI-2, material atmospheric harm; PI-3, passive-band protection; PI-4, spectrum coexistence; PI-5, treaty compliance; PI-6, national security coordination; PI-7, character qualification; PI-8, outstanding convergent disclosures; PI-9, institutional integrity.

Full development preserved.

Preservation Notice

The full substance of each gate, including record pin cites, governing authorities, independently-sufficient analysis for each claim, and the cross-reference matrix linking each claim to the relief waterfall, is preserved for any future proceeding. No argument is waived by its summary treatment in this appendix.

APPENDIX D: VIABLE ALTERNATIVE ARCHITECTURES

Purpose

The Application proposes one million LEO satellites for orbital computing, but the record does not explain why that architecture was chosen over alternatives that would achieve similar objectives with fewer regulatory, environmental, governance, and treaty-compliance concerns. Without that comparison, the Commission cannot make an informed public-interest finding under Section 309(a).

Viable alternatives include:

- Ground-based AI data centers within established regulatory frameworks;
- MEO architectures requiring far fewer satellites and permitting more feasible interference analysis;
- Lunar or cislunar computing that avoids LEO congestion and atmospheric reentry emissions;
- Hybrid architectures combining limited LEO relay or edge processing with ground-based computation;
- Phased deployment through a bounded initial system supported by later expansion filings; and
- Recovery-and-reuse and orbital foundry models that avoid routine atmospheric deposition from burn-up.

The Commission should ask why the Applicant chose the architecture that creates the greatest regulatory friction when less burdensome alternatives appear available. That question bears directly on the public-interest determination.